

Oluwatobi Melvyn Mayungbo

Machine Learning Engineer

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SUMMARY

Machine learning engineer with a B.Sc. in Statistics who reports precision, recall and baselines before trusting a model. Built fraud classifiers benchmarked across 3 seeds with a per-attack-type breakdown, an explainable risk scorer measured on hand-checked cases, and a reconciliation system with human-in-the-loop review.

SKILLS

Machine Learning: Scikit-learn, LightGBM, PyTorch, SHAP, Feature Engineering, Model Calibration (Isotonic Regression, Temperature Scaling), XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers)

Engineering and MLOps: Python, FastAPI, Docker, Pytest, CI/CD (GitHub Actions), SQL, DuckDB, PostgreSQL, Redis, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, Git, Streamlit, React

AI and LLM Systems: LLM Evaluation and Guardrails, Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

Payments and Risk: Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

PROJECTS

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds, with a per-attack-type breakdown covering card testing, account takeover and merchant compromise; sequence models cut expected cost per transaction 27% under an explicit cost model.
- Handled extreme class imbalance with focal loss and weighted sampling and isolated their failure modes: focal loss degraded probability calibration (ECE 0.0063) while weighted sampling cost 0.042 PR-AUC. Corrected calibration with temperature scaling and isotonic regression, reducing ECE to 0.0005.
- Deployed real-time scoring by exporting to ONNX Runtime (2.9x faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Designed a rule-based, fully explainable scorer where findings carry stable IDs and decisive findings set a floor of 75 so trust signals can never hide them; each score carries a confidence level based on how many sources answered.
- Assembled an evaluation set of 34 hand-checked addresses (12 risky, 22 safe) with a record and replay harness reporting ROC AUC, precision and recall.

Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL

- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action